A Pre-Deployment Audit for Conformal Coverage under Model Compression

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Abstract

We present a pre-deployment audit that predicts, from data already in hand, how much conformal coverage a classifier will lose when it is compressed. Deployed models are routinely quantized after calibration, and we show empirically that this silently invalidates the guarantee while accuracy is unchanged. The audit computes the shift that compression induces in the nonconformity-score distribution on the labeled calibration set and outputs a computable upper bound on the resulting coverage deficit; no new data, labels, or deployment access is required. Across 2 datasets $\times$ 2 architectures $\times$ 5 bit-widths (60 runs), the bound held in every run, and in the damage regime on peaked score distributions it is nearly tight: at 4-bit and 3-bit precision the measured deficit matched the total-variation distance to within 0.003. We further characterize what recalibration can and cannot repair: validity is restored at every precision tested, including models degraded to 47.8% accuracy, while informativeness collapses past an architecture-dependent precision knee (mean prediction-set size 0.91 at 8-bit versus 5.6 to 9.4 classes at 3-bit). A deployment corollary under simulated sensor drift shows that refreshing only the calibration scores (kilobytes; no retraining) maintains validity where launch calibration decays from 90.7% to 20.9% coverage, with prediction-set growth acting as the model-health signal. All results are reproducible from public code and raw data.

1 Introduction

Split conformal prediction wraps any classifier with a distribution-free guarantee: prediction sets that contain the true label with probability at least $1-\alpha$ under exchangeability [1, 2, 3]. The guarantee is calibrated on held-out data using the model's nonconformity scores. In deployment practice, however, the model that ships is rarely the model that was calibrated. Post-training quantization (PTQ), pruning, and distillation intervene between calibration and deployment, particularly in resource-constrained settings (mobile, embedded, onboard spacecraft) where compression is mandatory.

This paper asks a narrow question with practical consequences: *what does compression do to a conformal guarantee calibrated on the uncompressed model, and can the damage be predicted in advance?* Our answers:

1. **Stale calibration silently undercovers** (Section 4.1). Reusing FP32 calibration on an INT8 deployment undercovers at every miscoverage level tested, while accuracy is essentially unchanged, the failure is invisible to accuracy monitoring.

2. **The loss is forecastable** (Sections 3, 4.3). The coverage deficit is upper-bounded by the total-variation (TV) distance between reference and deployed score distributions on the calibration



Figure 1: The failure the audit predicts, on one real Sentinel-2 tile: prediction sets under a stale calibration threshold as weight precision drops.

set, a quantity computable on the ground, before deployment, with no new data. The bound is elementary; the contribution is empirical: it held in 60/60 runs across datasets, architectures, and bit-widths, and in the damage regime on peaked score distributions it is nearly an equality.

3. **Recalibration restores validity but not informativeness** (Section 4.4). Coverage returns to target at every precision, even for badly damaged models; mean prediction-set size, by contrast, exhibits a sharp knee in precision, one full bit earlier for MobileNetV3 than for ResNet18. Set size on calibration data is therefore the natural acceptance test for a compressed artifact.

4. **Deployment corollary** (Section 5): under progressive sensor drift, periodically refreshing the calibration scores (200 labeled samples per refresh; no retraining) largely preserves validity, and set-size growth signals when the model itself, rather than its calibration, requires intervention.

2 Related work

Conformal prediction under shift. Barber et al. [4] bound the coverage gap of conformal prediction beyond exchangeability by the total variation between calibration and test distributions, with weighted variants; subsequent work refines such bounds under Wasserstein and other metrics. Our setting is complementary: the data remain exchangeable, and the shift is induced in the *score function* by compressing the model. We instantiate the same style of bound for this self-inflicted, fully measurable shift and characterize its tightness empirically.

Uncertainty for onboard and remote-sensing models. Marchante Arjona et al. [5] study conformal prediction for onboard Earth observation under domain shift with weighted conformal methods, and the same group proposes class-conditional randomized smoothing against structured spectral corruptions [6]. These address input-side shift; neither considers compression of the model itself, which is the axis studied here. We view the two as composable layers of the same deployment problem.

Compression and uncertainty. A recent benchmark studies uncertainty under compression for large language models [7], reporting threshold-like degradation. Our vision results differ: on EuroSAT and CIFAR-10 the collapse of stale-calibration coverage is smooth and tracks the measured score shift (Section 4.2), and we additionally study the stale-versus-recalibrated distinction and set-size efficiency, which that benchmark does not address.

Protocol: pre-deployment coverage audit

Inputs: reference model f_{ref}, compressed candidate f_q, labeled calibration set D_{cal}, level α.

1. Score D_{cal} under f_{ref} and under f_q.
2. Estimate d_{TV} between the two score distributions (histogram).
3. Report forecast coverage floor $1 - \alpha - d_{TV}$ under stale calibration; if unacceptable, recalibrate $\hat{q}$ on the f_q scores.
4. Compute mean prediction-set size under the recalibrated threshold; reject the candidate if it exceeds the informativeness budget.
5. Ship f_q with its recalibrated threshold and the audit record.

Figure 2: The audit. One pass over the calibration set; no new data.

3 Setup and the forecasting bound

Let f_{ref} be the calibrated (reference) classifier and f_q its compressed deployment. Split conformal prediction with the LAC score $s(x, y) = 1 - p_y(x)$ computes the threshold $\hat{q}$ as the $\lceil (1-\alpha)(n+1) \rceil / n$ empirical quantile of calibration scores $S_i^{\text{ref}} = s_{\text{ref}}(X_i, Y_i)$, giving $\mathbb{P}(S^{\text{ref}} \leq \hat{q}) \geq 1 - \alpha$ on exchangeable data. If deployment instead evaluates f_q while retaining $\hat{q}$ (*stale calibration*), realized coverage is $\mathbb{P}(S^q \leq \hat{q})$, and the deficit obeys

$$(1 - \alpha) - \mathbb{P}(S^q \leq \hat{q}) \;\leq\; \sup_t |F_{S^{\text{ref}}}(t) - F_{S^q}(t)| \;=\; d_K(S^{\text{ref}}, S^q) \;\leq\; d_{TV}(S^{\text{ref}}, S^q), \tag{1}$$

up to the usual finite-sample slack in the calibration quantile. The inequality follows from a standard Kolmogorov/total-variation argument; its value here is operational. Because compression is applied by the deployer, both score distributions are computable on the *same labeled calibration set*: quantize, re-score, estimate d_{TV} (we use 1000-bin histograms on $[0, 1]$). The result is a pre-flight forecast: an upper bound on the coverage that will be lost if calibration is not refreshed. The empirical questions are whether the bound holds under estimation error and how tight it is; the remainder of the paper measures this.

4 Experiments

Protocol. Datasets: EuroSAT RGB [8] (27,000 tiles, 10 classes; 70/15/15 train/calibration/test) and CIFAR-10 [9]. Architectures: ResNet18 [10] and MobileNetV3-Small [11], ImageNet-initialized, 12 epochs AdamW. Compression: symmetric per-channel weight quantization to $b \in \{8, 6, 4, 3, 2\}$ bits (simulated PTQ: quantized values, FP32 arithmetic), plus production INT8 (FX-graph PTQ) for Section 4.1. Multi-seed throughout; error bars are standard deviations over seeds. Code, raw results, and exact environment are public; a fresh-clone reproduction test recovers the reported numbers exactly.[1]

4.1 Stale calibration silently undercovers

On EuroSAT/ResNet18 with production INT8 PTQ (20 seeds), accuracy is essentially unchanged (98.13% → 98.03%) while stale-calibration coverage falls below target at every level tested: 88.8±0.8%

[1] https://github.com/ss1738/euroSAT-CNN-landcover

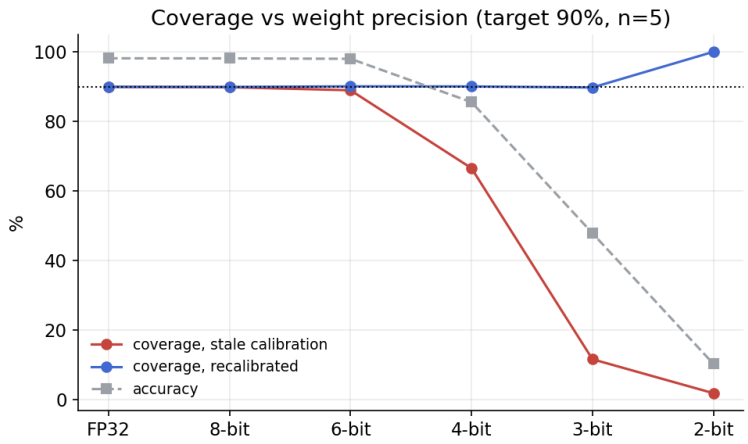


Figure 3: Coverage versus weight precision (EuroSAT/ResNet18, n=5 seeds, target 90%). Stale calibration (red) collapses smoothly; recalibration (blue) restores validity at every precision while accuracy (gray) falls.

against a 90% target (α=0.10), 94.4±0.6% against 95%, and 83.0±0.8% against 85%, the gap growing with α. Recalibrating on the quantized model restores target coverage at all levels (90.0 ± 0.7%, 94.9 ± 0.5%, 84.9 ± 0.6%). The practical rule is stark: *calibrate the artifact that ships, not the model that trained.*

4.2 Coverage collapse is smooth and tracks the score shift

Sweeping precision on EuroSAT/ResNet18 (5 seeds; Figure 3): stale-calibration coverage falls monotonically, 89.8% (8-bit), 88.9% (6-bit), 66.5% (4-bit), 11.6% (3-bit), 1.9% (2-bit) against a 90% target, tracking the relative weight perturbation (1.1%, 4.4%, 19.6%, 42.2%, 75.2%) and the mean absolute probability shift. We observe no threshold effect: degradation is gradual in the measured shift, in contrast to the threshold-like behaviour reported for LLMs [7]. Recalibration restores target coverage at *every* precision, including a 3-bit model at 47.8% accuracy (coverage 89.7%) and a 2-bit model at 10.3% accuracy (coverage 100%, attained by prediction sets growing toward all classes, see Section 4.4).

4.3 The bound holds and is nearly tight in the damage regime

Figure 4 plots measured deficit against estimated $d_{TV}(S^{\text{ref}}, S^q)$ per bit-width (5 seeds). At 8- and 6-bit the bound is conservative (deficit 0.002 and 0.011 against TV 0.030 and 0.034). In the damage regime it is nearly an equality: deficit 0.235 against TV 0.238 at 4-bit and 0.784 against 0.783 at 3-bit. The single nominal violation (0.001 at 3-bit) is within histogram-estimation error of the TV distance. Thus the forecast errs safe where damage is small and is near-exact where it matters.

Baselines and estimator choice. Table 2 compares pre-deployment predictors of the test deficit. The deficit measured directly on the calibration set is a near-exact point forecast (e.g. 0.234 vs. a true 0.235 at EuroSAT 4-bit) but is two-sided: it can and does underestimate (0.772 vs. 0.776 at 3-bit). The distributional distances are one-sided by Eq. (1): the Kolmogorov distance d_K is the tighter certified bound and never underestimated in any run, with d_{TV} more conservative; the accuracy drop badly underpredicts (0.123 vs. 0.235) and the Wasserstein distance carries no guarantee. The audit therefore reports the cal-shift as its point forecast and d_K as its certified

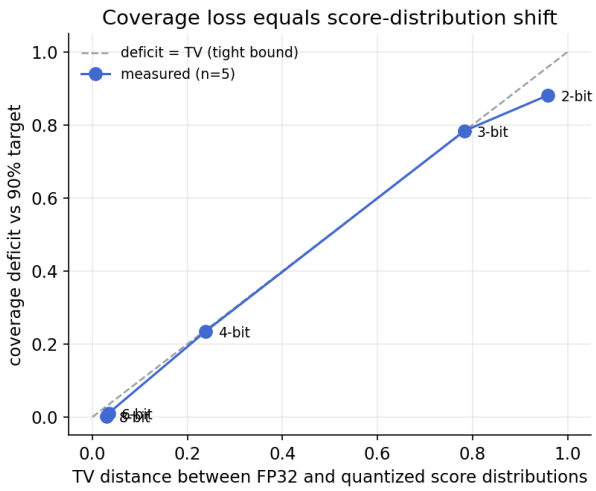


Figure 4: Coverage deficit versus TV distance between reference and deployed score distributions (EuroSAT/ResNet18). The dashed diagonal is deficit = TV.

floor. With the APS score in place of LAC, the bound held in all runs as well (stale-calibration deficits were near zero for APS, whose cumulative scores are less perturbed by quantization; largest observed deficit 0.003).

Generality. Repeating the test over the full grid (Table 1), {EuroSAT, CIFAR-10} × {ResNet18, MobileNetV3-Small} × {8,6,4,3,2}-bit, 3 seeds each, the bound of Eq. (1) held in **60 of 60 runs** (Figure 5). Tightness is systematic rather than accidental: on EuroSAT, whose score distribution is strongly peaked (high-confidence task), deficit approaches TV (e.g. 0.776 vs. 0.797, ResNet18 at 3-bit); on CIFAR-10, with broader score distributions, the bound is conservative (0.371 vs. 0.525). The forecast therefore never underestimated the loss in our experiments and approaches equality precisely on the peaked, high-accuracy regimes typical of deployed special-purpose classifiers. A secondary observation with engineering value: MobileNetV3 is markedly more quantization-fragile than ResNet18 (deficit 0.623 vs. 0.235 at 4-bit on EuroSAT), consistent with per-channel quantization of depthwise convolutions; architecture choice is an assurance decision, not only an accuracy one.

4.4 Validity is recoverable; informativeness is not

Recalibration always restores coverage, but the guarantee can be honored vacuously. Mean prediction-set size after recalibration (Figure 6) stays near one class down to 6-bit (0.91 to 1.24 across the grid), passes a knee at 4-bit (1.14 for EuroSAT/ResNet18 but already 4.94 for EuroSAT/MobileNetV3), and explodes at 3-bit (5.2 to 9.4 classes), saturating near all 10 classes at 2-bit. Validity and informativeness thus separate cleanly under compression: the former is always repairable by re-scoring, the latter has an architecture-dependent precision floor. Mean set size on the calibration set is consequently a one-line acceptance test for whether a compressed artifact is worth deploying.

5 Deployment corollary: refresh the guarantee, not the model

Compression is one source of score shift; sensor drift over a mission is another, and the same machinery applies. Under a simulated 24-week progressive degradation (additive noise ramping to σ=0.05 and brightness to ×1.35; EuroSAT/ResNet18, 3 seeds, Figure 7), never-refreshed launch

Table 1: Coverage deficit vs. the TV forecast across the full grid (α=0.10, n=3 seeds, mean±sd). Ratio = deficit/TV; the bound holds when ratio ≤ 1. It held in all 60 individual runs.

Dataset	Model	Bits	Deficit	TV forecast	Ratio
EuroSAT	ResNet18	8	-0.002±0.002	0.054±0.002	-0.04
EuroSAT	ResNet18	6	0.007±0.010	0.063±0.013	0.11
EuroSAT	ResNet18	4	0.235±0.013	0.290±0.021	0.81
EuroSAT	ResNet18	3	0.776±0.050	0.797±0.036	0.97
EuroSAT	ResNet18	2	0.869±0.054	0.955±0.043	0.91
EuroSAT	MobileNetV3-S	8	0.016±0.009	0.121±0.021	0.13
EuroSAT	MobileNetV3-S	6	0.121±0.084	0.271±0.056	0.45
EuroSAT	MobileNetV3-S	4	0.623±0.055	0.698±0.041	0.89
EuroSAT	MobileNetV3-S	3	0.870±0.014	0.925±0.013	0.94
EuroSAT	MobileNetV3-S	2	0.805±0.011	0.902±0.011	0.89
CIFAR-10	ResNet18	8	-0.001±0.001	0.138±0.004	-0.00
CIFAR-10	ResNet18	6	0.002±0.001	0.139±0.004	0.02
CIFAR-10	ResNet18	4	0.072±0.044	0.247±0.046	0.29
CIFAR-10	ResNet18	3	0.371±0.090	0.525±0.036	0.71
CIFAR-10	ResNet18	2	0.667±0.049	0.822±0.042	0.81
CIFAR-10	MobileNetV3-S	8	-0.000±0.004	0.192±0.005	-0.00
CIFAR-10	MobileNetV3-S	6	0.018±0.003	0.193±0.005	0.09
CIFAR-10	MobileNetV3-S	4	0.142±0.059	0.310±0.019	0.46
CIFAR-10	MobileNetV3-S	3	0.468±0.006	0.556±0.008	0.84
CIFAR-10	MobileNetV3-S	2	0.797±0.003	0.985±0.009	0.81

calibration decays from 90.7% to 20.9% coverage. Refreshing *only the conformal scores* every 4 weeks, 200 labeled samples per refresh, kilobytes of storage, no retraining or gradients, holds coverage at 81 to 87% through end of test; a 12-week cadence is insufficient (54.0% at week 23), so refresh rate must track drift rate. Under the working refresh, mean set size grows from 0.91 to 6.65 as model accuracy falls from 97.8% to 30.9%: the guarantee remains honest by becoming less informative, and set-size growth is precisely the onboard signal that the model, not the calibration, now requires intervention.

6 Limitations

The bound in Eq. (1) is one-directional and follows from standard arguments; the claims of this paper are the audit protocol, its measured tightness regime, and its validation, not new theory. TV is estimated by histograms; at extreme damage (3-bit) the estimator error is visible as a 0.001 nominal violation. Experiments cover split conformal classification with the LAC score on two mid-scale vision datasets and two architectures; regression, other scores, and larger-scale models are untested. Quantization is simulated PTQ (quantized weights, FP32 arithmetic) except where production INT8 is stated; activation quantization is not studied. Drift is synthetic and the refresh policy assumes access to small labeled batches. Findings on fault injection and input-side monitoring from the same research programme are reported in the repository rather than here.

Table 2: Predicting the test-set coverage deficit before deployment (ResNet18, α=0.10, n=3 seeds). Cal-shift is the deficit measured directly on the calibration set (point forecast); d_K and d_{TV} are certified one-sided bounds (Eq. 1); accuracy drop and W_1 carry no guarantee and can underestimate.

Setting	True deficit	Cal-shift	d_K	d_{TV}	Acc. drop	W_1
EuroSAT, 8-bit	-0.002	-0.000	0.004	0.052	0.000	0.001
EuroSAT, 4-bit	0.235	0.234	0.274	0.288	0.123	0.128
EuroSAT, 3-bit	0.776	0.772	0.780	0.790	0.513	0.535
CIFAR-10, 8-bit	0.000	0.000	0.005	0.139	0.001	0.001
CIFAR-10, 4-bit	0.054	0.054	0.127	0.220	0.071	0.079
CIFAR-10, 3-bit	0.312	0.315	0.485	0.532	0.381	0.391

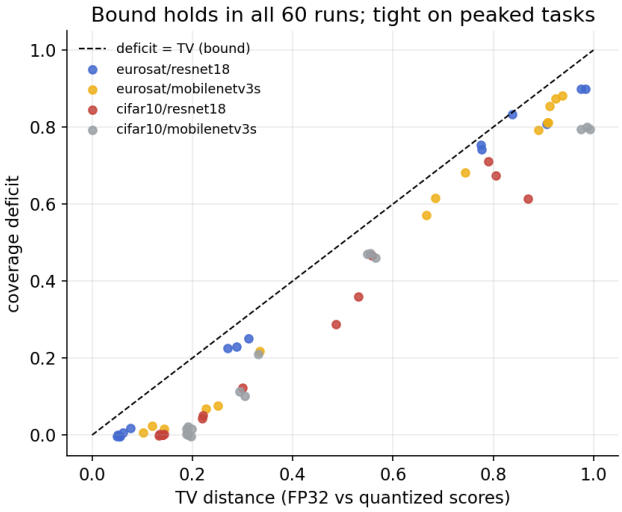


Figure 5: All 60 runs across the dataset × architecture × precision grid. Every point lies on or below the diagonal (bound holds); peaked-task points hug it.

7 Conclusion

A conformal guarantee does not survive model compression by default, but its loss is not a surprise to be discovered in deployment: it is a number that can be computed on the ground, from data already in hand, before the artifact ships. Together with the recoverability of validity, the precision knee in informativeness, and score-refresh as a maintenance policy, this yields a practical qualification recipe for compressed classifiers under distribution-free guarantees.

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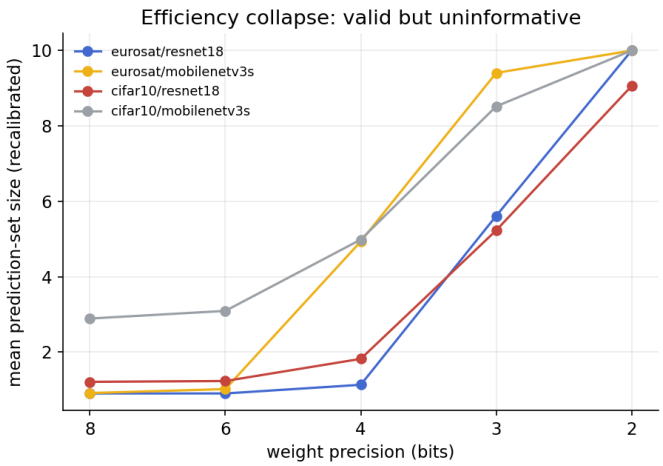


Figure 6: Mean recalibrated prediction-set size versus precision across the grid: valid but uninformative below the knee.

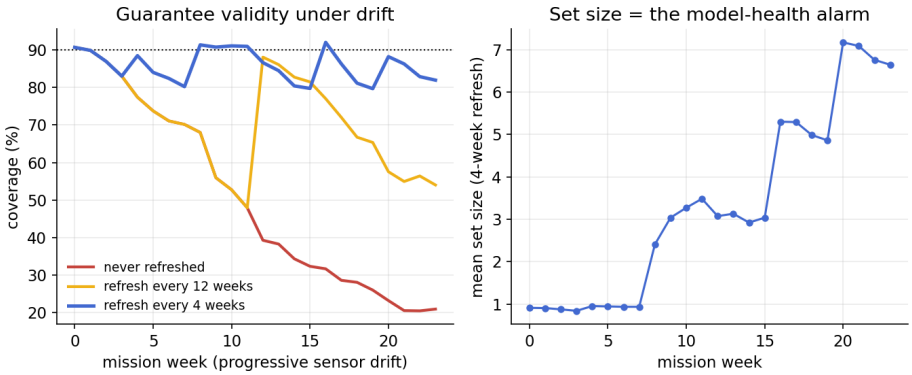


Figure 7: Simulated mission drift. Left: coverage under three maintenance policies (target 90%). Right: set size under the working policy is the model-health signal.

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